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Abstract

Reduced-order models of latent heat thermal energy storage commonly advance the melt front as an *area* or a *thickness*, driven by the heat that a resistance network delivers. Such a state variable cannot represent a control volume whose phase change is complete: the model must either melt material that is not there, or discard the delivered heat and break its own energy balance. In a finned tube exchanger with an axially graded melting temperature the second failure is not marginal — in the case examined here the discarded heat reaches 148 % of the heat delivered during discharge.

This paper formulates the segment state as an *enthalpy* per unit tube length measured from a fully-solid datum, so that subcooled solid, two-phase and superheated liquid are three branches of one single-valued curve and the melted fraction is confined to $[0, 1]$ by construction rather than by a guard. The segment equation is derived from energy balances on the phase-change material and on the fluid, and the closing conductance is shown to be $K = \dot{m}c_p(1 - e^{-\mathrm{NTU}})/\Delta z$ rather than the overall coefficient times the perimeter; the two differ by up to 33 % in the case studied, and only the former carries the capacity-rate limit that prevents heat flowing from cold to hot as the wall conductance improves.

The resulting equation is nonlinear but affine within each branch. A branchwise-exact integrator is presented that solves each branch in closed form and splits the step at branch crossings. It is unconditionally stable, preserves energy closure to machine precision by construction, and agrees with a 200000-substep explicit reference to 3×10^{-13}. Explicit stepping is shown to be inadmissible once sensible capacity is present, although it is exact on the latent plateau — which is why area-based formulations never exposed the constraint.

The formulation is demonstrated on a phase-change material packed into the annulus of a repurposed oil-and-gas well. Sensible capacity proves to be self-levelling: a control volume that finishes melting begins to superheat, its driving temperature difference collapses, and the heat passes downstream. Compared at equal exchanger size, the spread in local melt fraction falls from 0.405 to 0.000, and the phase-change material that never cycles — dead volume, carrying no energy through the round trip — falls correspondingly.

Finally, the formulation is carried through repeated cycles to a cyclic steady state. Because the state returns to itself and the store is adiabatic, the ratio of recovered to stored energy is identically unity at cyclic steady state, which removes the freedom that a single-cycle energy balance uses to size the exchanger. The sizing problem is therefore reformulated as a simulation: the hardware is specified, the cycle is marched to convergence, and the deviation of delivered energy from target is reported as a result rather than driven to zero.

Keywords: latent heat storage ; phase change material ; enthalpy method ; finned tube heat exchanger ; effectiveness–NTU ; exponential integrator

1 Introduction

Latent heat thermal energy storage couples a heat transfer fluid to a phase-change material (PCM) through a wall, and the engineering problem is almost always the same: the PCM conducts poorly, so the exchanger is finned, staged, or otherwise elaborated until the phase change can be driven at the required rate. Designing such an exchanger requires a model that couples a *heat transfer network* — convection, wall, fins, and a growing melt layer — to a *moving phase boundary*, over a duty cycle, at a cost low enough to permit parametric study.

Two families of reduced-order model are in common use. Closed-form Stefan solutions give the front position directly from a prescribed surface temperature, and are attractive because they are explicit. Energy-balance formulations instead advance the front from the heat that the network actually delivers, which enforces conservation between the two sides of the wall. This paper is concerned with a defect shared by both, and with what has to change to remove it.

1.1 The state variable, and what it cannot say

Both families carry the melt front as a geometric quantity: a layer thickness δ, or equivalently a melted cross-sectional area A_{melt}. Ask such a model for the temperature of the PCM in a given control volume and the only answer available is the melting temperature T_m, because that is all the state encodes. While solid remains, the answer is correct, which is why the limitation is easy to miss.

It stops being correct the moment a control volume completes its phase change. The network still reports a positive heat flow — the fluid is still hotter than T_m, and nothing in the resistance chain knows the PCM is exhausted — and the model must do one of two things:

1. **Continue melting.** A_{melt} grows past the PCM the control volume contains. Latent heat is drawn from material that is not there, and local melted fractions exceed unity.
2. **Clip and discard.** Cap A_{melt} at the available inventory and throw the surplus away. The melted fraction is then physical, but energy is not conserved, and the fluid leaves at a temperature that reflects heat which went nowhere.

Neither is acceptable, and no refinement of the marching scheme repairs either, because the deficiency is in the state: a variable counting melted area has no slot for liquid above T_m or solid below it. The remedy adopted here is to integrate enthalpy and recover the melted area from it — the classical enthalpy method [1, 2], applied not to a discretised field but to the lumped state of a one-dimensional exchanger segment.

1.2 Contribution

This paper makes three contributions, all concerning the formulation and its numerics rather than the application:

1. **A lumped enthalpy state for exchanger segments** (§2). One scalar per segment encodes subcooled solid, two-phase and superheated liquid; the melted fraction is confined to $[0, 1]$ by the constitutive relation rather than by a guard, and no heat is ever discarded.
2. **The correct closing conductance** (§3). Deriving the segment equation from control-volume balances yields $K = \dot{m}c_p(1 - e^{-\mathrm{NTU}})/\Delta z$, not $U_i P_i$. The two agree only as $\mathrm{NTU} \to 0$; the naive form lacks the capacity-rate limit and therefore permits unphysical behaviour as the wall conductance improves.
3. **A branchwise-exact integrator** (§4). Explicit stepping is exact on the latent plateau and inadmissible off it. The scheme presented solves each branch in closed form, splits steps at branch crossings, is unconditionally stable, and makes energy closure structural.

Section 5 demonstrates the formulation on a PCM-filled wellbore heat exchanger and reports the physical consequence that motivated the work: the self-levelling of the melt distribution.

2 Enthalpy formulation

2.1 Configuration and hypotheses

Consider a tube of developed length L_{tube} carrying a single-phase fluid at mass flow $\dot{m}$, surrounded by PCM. The tube is divided into n_s segments of length Δz along the developed coordinate s. Each segment owns a PCM cross-section A_{avail}, and the melting temperature may vary from segment to segment, permitting an axially graded (cascaded) store in which T_m follows the fluid glide.

The following are assumed throughout.

H1 Quasi-steady conduction in the melt layer: its temperature profile is the steady cylindrical profile for the instantaneous front position. The hypothesis is controlled by the Stefan number $\text{Ste} = c_p \Delta T / h_m$.

H2 No axial conduction in the wall or the PCM; segments couple only through the fluid stream.

H3 The melt region is a concentric annulus of uniform thickness δ about each tube, and fronts about neighbouring tubes do not interact.

H4 Conduction only in the melt; no natural convection in the liquid PCM.

H5 One PCM temperature per segment: the state is lumped, with no radial resolution of temperature *within* a phase.

H6 Adiabatic outer boundary.

H7 Incompressible single-phase fluid, treated as quasi-steady.

2.2 Why enthalpy rather than temperature

The natural repair for the deficiency of §1 is to carry a PCM temperature alongside the melted area. It is the wrong repair, for a reason that recurs throughout phase-change modelling: at a phase change, temperature is not invertible. During melting the PCM remains at T_m while its energy content climbs by the entire latent heat, so T carries no information whatever throughout the process that dominates the problem. A pair (T, A_{melt}) works, but obliges the code to decide at every step which member is active, keep the two consistent, and handle the instants at which the active member changes.

Enthalpy is invertible. Every state of the PCM corresponds to exactly one value of the enthalpy, and the enthalpy increases monotonically as heat is added. A single scalar per segment therefore encodes all three regimes, and the regime is read from the value rather than tracked separately (Fig. 1).

2.3 State and capacities

Let E' be the enthalpy per unit tube length of the PCM belonging to one segment, measured from a datum of fully solid material at T_m. The datum is arbitrary; this choice places the two branch boundaries at 0 and E'_{lat}, which shortens every test in the implementation. With

$$E'_{\text{lat}} = \rho_m h_m A_{\text{avail}}, \qquad C_s = \rho_m c_{p,s} A_{\text{avail}}, \qquad C_l = \rho_m c_{p,l} A_{\text{avail}}, \tag{1}$$

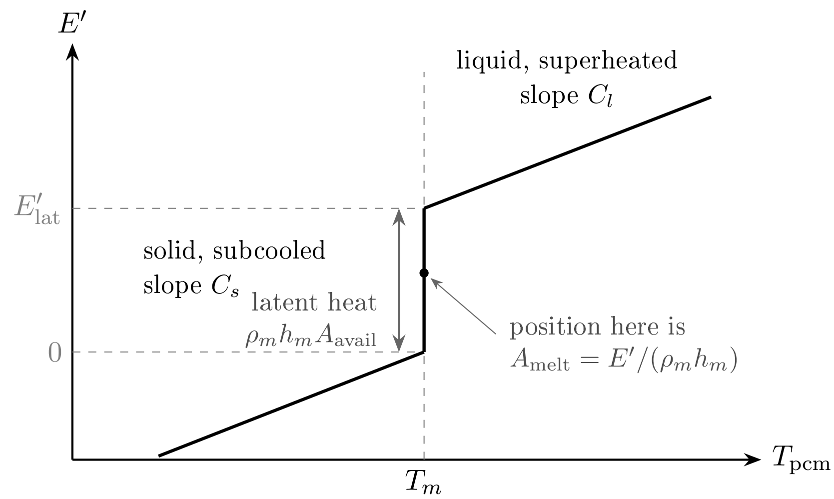

Figure 1: The constitutive curve, Eq. (2). Read as $E'(T)$ it is multivalued at T_m, which is why temperature cannot serve as the state. Read as $T(E')$ — the direction used here — it is single-valued everywhere, and position along the vertical segment is the melted fraction.

the latent capacity and the two sensible capacities per unit tube length, the constitutive curve and the melted area are

$$T_{\text{pcm}} = \begin{cases} T_m + E'/C_s, & E' < 0 \quad \text{(solid, subcooled)}, \\ T_m, & 0 \le E' \le E'_{\text{lat}} \quad \text{(two-phase)}, \\ T_m + \big(E' - E'_{\text{lat}}\big)/C_l, & E' > E'_{\text{lat}} \quad \text{(liquid, superheated)}, \end{cases} \tag{2}$$

$$A_{\text{melt}} = \frac{\min\!\big(\max(E', 0),\, E'_{\text{lat}}\big)}{\rho_m h_m}, \qquad \varepsilon_{\text{local}} = \frac{A_{\text{melt}}}{A_{\text{avail}}} \in [0, 1]. \tag{3}$$

The saturation in Eq. (3) is not a guard added to suppress bad behaviour; it is the definition. Melted area saturates because a segment cannot melt PCM it does not own, and superheat is where the additional energy goes. Consequently $\varepsilon_{\text{local}} \in [0, 1]$ is *structural*: no code path can violate it, and no test for it is required.

2.4 Recovering the layer thickness

The resistance network requires δ, not A_{melt}. For a tube of outer radius r_e carrying n_f longitudinal fins of thickness t_f and radial length L_f, the fin metal occupies part of the annulus and must be excluded from the melted PCM area:

$$A_{\text{melt}} = \pi \left[(r_e + \delta)^2 - r_e^2\right] - n_f t_f \min(\delta, L_f)\,, \tag{4}$$

which inverts in closed form on each branch,

$$\delta \le L_f: \quad \pi\delta^2 + (2\pi r_e - n_f t_f)\,\delta - A_{\text{melt}} = 0, \tag{5}$$

$$\delta > L_f: \quad \pi\delta^2 + 2\pi r_e \delta - (A_{\text{melt}} + n_f t_f L_f) = 0. \tag{6}$$

For $n_f = 0$ both reduce to $\delta = \sqrt{r_e^2 + A_{\text{melt}}/\pi} - r_e$.

The melt layer then enters the network as a cylindrical conduction shell,

$$h_e = \frac{k_m}{r_e \ln(1 + \delta/r_e)}, \tag{7}$$

and the overall coefficient on the inner area, with fin efficiency η_f, overall surface efficiency η_o and finned perimeter P_T, is

$$U_i = \left[\frac{1}{h_i} + R''_{f,i} + \frac{r_i}{k_w} \ln \frac{r_e}{r_i} + \frac{2\pi r_i}{P_T\, \eta_o\, h_e} \right]^{-1}. \tag{8}$$

The single change relative to an area-based formulation is that the outer boundary of this chain sits at T_{pcm}, not at T_m. It coincides with T_m only while solid remains. Section 5 shows that this one substitution is responsible for the principal physical result.

3 The segment equation and its closure

Equation (14) below is short enough to look like a definition. It is not: it follows from energy balances on two control volumes and the elimination of the quantity they share. The derivation is given in full because the closing conductance is not the one a reader would write down unaided.

3.1 Two control volumes

Fix a segment j spanning $s \in [s_j, s_{j+1}]$, $\Delta z = s_{j+1} - s_j$. Two control volumes share the tube wall over that interval: **CV 1**, the PCM belonging to the segment — cross-section A_{avail}, stationary, closed — and **CV 2**, the fluid within the tube over the same interval, open, entering at T_j and leaving at T_{j+1}. Write q'_j for the heat per unit tube length crossing from CV 2 into CV 1.

3.2 The PCM balance

CV 1 is closed and stationary, so the first law reduces to a statement about its enthalpy. With the axial (H2) and outer-boundary (H6) terms dropped,

$$\boxed{\frac{\mathrm{d}E'_j}{\mathrm{d}t} = q'_j} \tag{9}$$

3.3 The fluid balance

For incompressible single-phase flow without viscous dissipation, the energy equation per unit tube length is

$$\rho A_{\text{flow}} c_p \frac{\partial T}{\partial t} + \dot{m} c_p \frac{\partial T}{\partial s} = -q'(s). \tag{10}$$

Hypothesis H7 drops the first term. The approximation is justified by the separation between the fluid transit time and the timescale of the PCM state, and its cost is an unresolved startup transient of one transit at the beginning of each half-cycle; §5 quantifies both for the case studied.

With CV 1 presenting a single temperature to the whole segment (H5), the local flux closes on the network,

$$q'(s) = U_i\, (2\pi r_i)\, [T(s) - T_{\text{pcm}}]\,. \tag{11}$$

The inner perimeter $2\pi r_i$ appears rather than P_T because U_i was referred to the inner area in Eq. (8); the finned perimeter is already inside it. Using P_T here would count the fins twice.

3.4 Integration across the segment

Substituting Eq. (11) into the steady form of Eq. (10), with T_{pcm} and U_i held constant over the segment, gives a linear ordinary differential equation in s whose solution is the single-stream heat exchanger result:

$$T_{j+1} = T_{\mathrm{pcm}} + (T_j - T_{\mathrm{pcm}})\, e^{-\mathrm{NTU}_j}, \qquad \mathrm{NTU}_j = \frac{2\pi r_i\, U_i\, \Delta z}{\dot{m}\, c_p}. \tag{12}$$

One stream carries a finite capacity rate $\dot{m}c_p$; the other is an isothermal reservoir of infinite capacity rate, with effectiveness $\varepsilon = 1 - e^{-\mathrm{NTU}}$. The reservoir idealisation is precisely what the lumped state of H5 licenses.

3.5 The closure, and why the conductance is not $U_i P_i$

The heat the fluid gives up over the segment is the enthalpy it loses, so from Eq. (12),

$$\boxed{q'_j = K_j\,(T_j - T_{\mathrm{pcm}})\,, \qquad K_j = \frac{\dot{m}c_p\left(1 - e^{-\mathrm{NTU}_j}\right)}{\Delta z}.} \tag{13}$$

Combining Eqs. (9), (13) and (2) closes the system:

$$\boxed{\frac{\mathrm{d}E'}{\mathrm{d}t} = K\left(T_j - T_{\mathrm{pcm}}(E')\right)} \tag{14}$$

with one unknown per segment; everything else is geometry, a property, or the upstream fluid temperature handed down by the preceding segment.

The naive closure evaluates Eq. (11) at $T(s) \approx T_j$, giving $q'_j \approx 2\pi r_i U_i (T_j - T_{\mathrm{pcm}})$. Comparing,

$$\frac{K_j}{2\pi r_i U_i} = \frac{1 - e^{-\mathrm{NTU}_j}}{\mathrm{NTU}_j} \le 1, \tag{15}$$

and the two limits say what the difference means. As $\mathrm{NTU} \to 0$ the ratio tends to unity: the fluid barely cools across the segment and the naive closure is correct. As $\mathrm{NTU} \to \infty$, $K \to \dot{m}c_p/\Delta z$, a finite limit set by the *fluid* rather than the wall — a segment cannot absorb more than the stream delivers to it. The naive closure has no such limit: it permits T_{j+1} to fall below T_{pcm}, that is, heat flowing from cold to hot, at a rate growing without bound as U_i improves. The discrepancy is largest where the melt layer is thinnest and the exchanger best, which is where the model is most active.

3.6 Closure is structural

Equation (13) is not evaluated independently of Eq. (9). The implementation integrates Eq. (9) over the step and *reports*

$$q'_j = \frac{E'^{\,n+1} - E'^{\,n}}{\Delta t}, \tag{16}$$

then drops the fluid temperature by exactly that amount, $T_{j+1} = T_j - q'_j \Delta z/(\dot{m}c_p)$. Whatever the PCM gained, the fluid lost, to machine precision, with no possibility of drift. It should be stressed that this is a *consistency* property and not evidence of correctness: the two operations are inverse, so a small residual measures floating-point arithmetic and nothing else.

4 Branchwise-exact time integration

4.1 Why explicit stepping is inadmissible

Equation (14) is nonlinear, because $T_{\mathrm{pcm}}(E')$ is piecewise, but affine within each branch of Eq. (2). The distinction governs the choice of integrator.

On the latent plateau $T_{\mathrm{pcm}} = T_m$ is constant and does not respond to E' at all. The segment behaves as though its heat capacity were infinite, Eq. (14) reduces to $\mathrm{d}E'/\mathrm{d}t = \mathrm{const}$, and an explicit Euler step is not merely stable but exact. This is the regime an area-based formulation occupies permanently, which explains why the constraint derived below has not previously been a concern: explicit stepping was safe for a structural reason, not a fortunate one.

Introducing sensible capacity makes the capacity finite. Equation (14) becomes a relaxation equation with time constant C/K, and explicit Euler requires $\Delta t < C/K$. Logarithmic time grids, which are the natural choice for a diffusion-limited process, violate this at late times by a wide margin; §5 gives the figures for the case studied, where the final step exceeds the limit by a factor of 7.7 and an explicit implementation diverges.

4.2 The scheme

Rather than reduce Δt — which for a logarithmic grid multiplies the cost by some three orders of magnitude — the linearity within each branch is exploited and each branch integrated in closed form. Holding T_j fixed across the step,

$$E'(t+\Delta t) = \begin{cases} E' + K\,(T_j - T_m)\,\Delta t, & \text{plateau,} \\ E'_{\mathrm{eq}} + \big(E' - E'_{\mathrm{eq}}\big)e^{-K\Delta t/C}, & \text{sensible branches,} \end{cases} \tag{17}$$

with the equilibrium enthalpy and relevant capacity

$$\big(C, E'_{\mathrm{eq}}\big) = \begin{cases} \left(C_s,\ C_s\,(T_j - T_m)\right), & E' < 0, \\ \left(C_l,\ E'_{\mathrm{lat}} + C_l\,(T_j - T_m)\right), & E' > E'_{\mathrm{lat}}. \end{cases} \tag{18}$$

E'_{eq} is the enthalpy at which the PCM would sit in equilibrium with the fluid: the state the segment relaxes toward and never reaches. The exponential is unconditionally stable and monotone, so no step, however large, can overshoot it. That is precisely the property explicit Euler lacks.

4.3 Branch crossings

A step may begin in one branch and end in another, in which case it is split at the crossing. With E'_b the boundary being approached (0 or E'_{lat}), the crossing time is

$$t^* = \begin{cases} \dfrac{E'_b - E'}{K\,(T_j - T_m)}, & \text{from the plateau,} \\[2ex] \dfrac{C}{K}\ln\!\left(\dfrac{E' - E'_{\mathrm{eq}}}{E'_b - E'_{\mathrm{eq}}}\right), & \text{from a sensible branch.} \end{cases} \tag{19}$$

If $t^* < \Delta t$ the state is advanced to E'_b, the remaining $\Delta t - t^*$ is integrated in the adjacent branch, and the test repeated. Only a small number of crossings can occur within one step, so the loop is bounded.

Table 1: Algorithm: advance one segment over Δt.

1	$t_{\text{left}} \leftarrow \Delta t, \quad E'_0 \leftarrow E'$
2	**while** $t_{\text{left}} > 0$:
3	identify the branch containing E'
4	compute t^* from Eq. (19) toward the branch boundary
5	**if** $t^* \geq t_{\text{left}}$: advance by Eq. (17) over t_{left}; **break**
6	**else**: set $E' \leftarrow E'_b \pm \epsilon; \quad t_{\text{left}} \leftarrow t_{\text{left}} - t^*$
7	**return** $E', \quad q' = (E' - E'_0)/\Delta t$

A note on the boundary test. Step 6 of Table 1 displaces the state just past the boundary by $\epsilon = 10^{-9} \max(E'_{\text{lat}}, 1)$ rather than placing it exactly on the boundary. This is necessary rather than cosmetic. If the branch test uses closed intervals, a state landing exactly on E'_{lat} is still classified as two-phase; the plateau branch then computes $t^* = 0$, advances the state to where it already is, subtracts zero from the remaining time, and repeats indefinitely. The segment is pinned at the boundary and never enters the liquid branch. The symptom is not a failure but an apparent success — every segment reporting a melted fraction of exactly unity and zero superheat. The general point is that where a solver switches on the sign of a quantity, the side of the boundary to which equality belongs must be chosen deliberately, and the state must not be able to come to rest there.

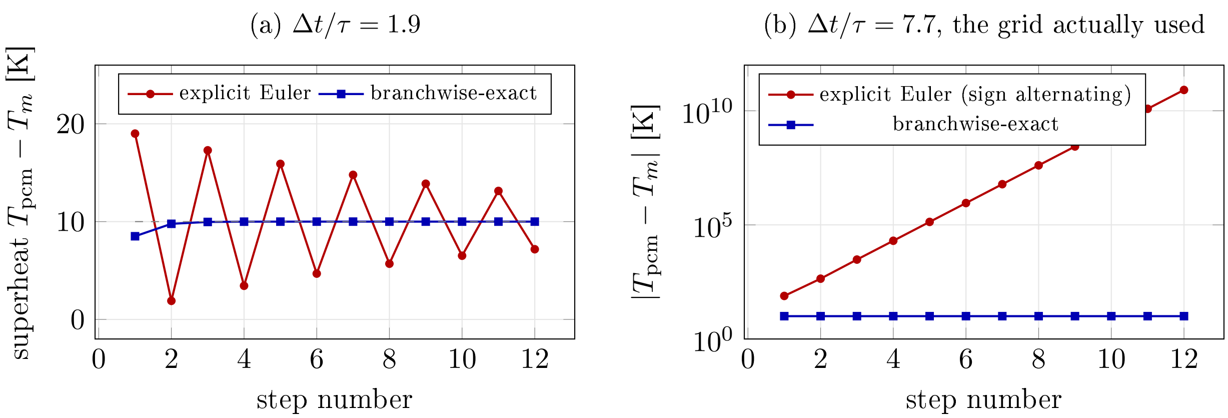

Figure 2: Response of a single segment on the liquid branch, driven by fluid 10 K above T_m with $\tau = C_l/K = 1102$ s. (a) At $\Delta t/\tau = 1.9$ explicit Euler oscillates about the equilibrium while remaining bounded. (b) At $\Delta t/\tau = 7.7$ — the ratio the logarithmic grid of Table 4 actually reaches at the end of a charge — it diverges, alternating in sign and passing 8e10 K within twelve steps. The branchwise-exact scheme is monotone and correct at every step size, because Eq. (17) cannot overshoot E'_{eq}.

4.4 Verification

The scheme was verified against a brute-force explicit integration of Eq. (14) using 200000 substeps, over a complete charge and discharge, including steps crossing branch boundaries. Agreement is 3×10^{-13} in relative terms, which is machine precision. The two calculations share no code path beyond Eq. (2), so this is a genuine verification of the integrator rather than a consistency check.

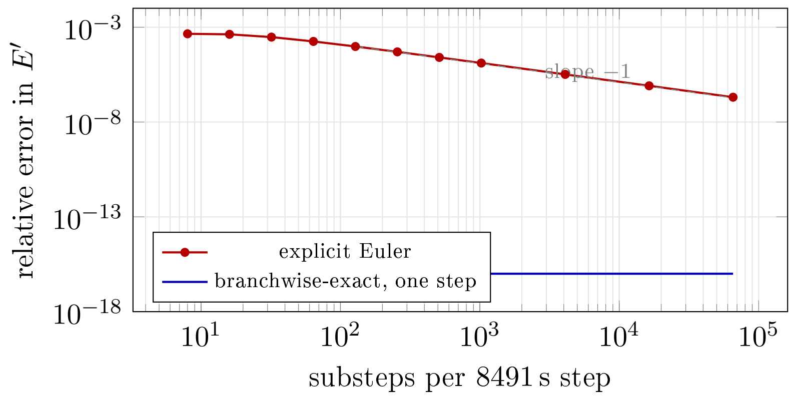

Figure 3: Error after advancing one segment through a single 8491 s grid step, against the number of explicit substeps used. Explicit Euler is first order once it is stable (≥ 8 substeps; below that it is not), and needs of order 10^4 substeps to reach 10^{-6}. The branchwise-exact scheme returns the closed-form solution of the branch, so its error is zero to machine precision in a *single* step — plotted at 10^{-16} for visibility.

Energy closure, by contrast, is structural under Eq. (16) and therefore proves nothing about correctness; it is reported here only to establish that no drift is present.

5 Demonstration: a PCM-filled wellbore heat exchanger

5.1 Configuration

The formulation is demonstrated on latent heat storage in the annulus of a repurposed oil-and-gas well. A high-temperature heat pump melts the PCM during charging and an organic Rankine cycle recovers the energy during discharging, with pressurised water circulating through finned hairpin tubes in the borehole. Table 2 gives the configuration.
The resulting capacities, Eq. (1), are $A_{\text{avail}} = 4.5411\text{e} - 3\,\text{m}^2$, $E'_{\text{lat}} = 2.675\,\text{MJ/m}$, $C_s = 9.010\text{e}3\,\text{J/m/K}$ and $C_l = 1.267\text{e}4\,\text{J/m/K}$. The ratio $C_l/E'_{\text{lat}} = 4.74\text{e} - 3\,/\text{K}$ is the Stefan number in disguise: one kelvin of superheat is worth 0.47 % of the latent capacity, so with the 10 K approach available here the sensible branches carry at most about 4.7 % of the energy. A reader might reasonably conclude that the formulation is a small correction. Section 5.5 shows why it is not.

5.2 Quasi-steady fluid: the cost of H7

Table 3 collects the comparisons that justify dropping $\partial T/\partial t$ from Eq. (10).
The fluid re-establishes its profile in one transit while the PCM state evolves over ten hours, a separation of about sixteen. The approximation is therefore reasonable but not asymptotic: the first 38 min of each half-cycle is a startup transient the model does not resolve, some 6 % of the window. What renders this tolerable is the last row — the energy held in the fluid is only 1.2 % of the latent inventory it is charging, so the transient misplaces very little energy even though it occupies a noticeable fraction of the run.

Table 2: Configuration of the demonstration case.

	quantity	value	unit
Borehole	depth L_{well}	1524	m
	diameter D_{well}	177.8	mm
	hairpin tubes n_t	2	–
	PCM volume V_{well}	27.68	m^3
Tube	inner radius r_i	16.23	mm
	outer radius r_e	21.08	mm
	fins per leg n_f	24	–
	fin $L_f \times t_f$	7.5×1.5	mm
PCM	melting temperature T_m	150	°C
	latent heat h_m	380	kJ/kg
	$c_{p,s}$ / $c_{p,l}$	1280 / 1800	J/kg/K
	k_s / k_l	0.60 / 0.45	W/m/K
	ρ_s / ρ_l	1550 / 1450	kg/m^3
Operation	fluid glide $\Delta T_{3C,2C}$	55	K
	charging inlet	160	°C
	charge / discharge window	10 / 10	h
	PCM layers N_{lay}	9	–

Table 3: Justification and cost of the quasi-steady fluid approximation.

quantity	value	
fluid transit time over L_{tube}	2264 s	($u = 1.35$ m/s)
charging window t_{ch}	36000 s	
ratio	0.063	
$\rho A_{\text{flow}} c_p / C_l$	0.257	fluid vs. PCM sensible capacity
$\rho A_{\text{flow}} c_p \Delta T / E'_{\text{lat}}$	0.0122	fluid swing vs. latent, $\Delta T = 10$ K

5.3 The closure discrepancy in practice

Over the charge, NTU ranges from 0.083 to 0.602, so Eq. (15) lies between 0.96 and 0.75. Using U_iP_i in place of Eq. (13) would therefore overstate the segment conductance by between 4 % and 33 % in this case, with the largest error at early times when the melt layer is thin and the exchanger at its best.

5.4 Stability of the time integration

Table 4 gives the stability figures for the logarithmic time grid used here.

Table 4: Explicit-Euler stability limit against the time grid actually used, for the first segment.

	early ($t = 1\,\mathrm{s}$)	late ($t \to t_{\mathrm{ch}}$)
NTU	0.592	0.083
K [W/m/K]	64.3	11.5
C_l/K [s]	197	1106
grid step Δt [s]	0.31	8491
$\Delta t\, K/C_l$	0.002	7.7

The grid is comfortable at the start and violates the limit by almost an order of magnitude at the end. An explicit implementation duly diverged, the oscillation feeding back through the fluid temperature and driving the secondary fluid to 261 K — some 170 K below anything physical. That failure was loud and immediately diagnosable; the dangerous case is a step ratio nearer 1.5, unstable in principle but slowly enough to resemble a plausible answer.

5.5 Sensible capacity is self-levelling

The principal physical result is not the magnitude of the sensible energy but its effect on the *distribution* of melting.

Consider the segment nearest the inlet, which sees the hottest fluid and therefore melts first. Under an area-based formulation it finishes melting and then continues to absorb heat at the full driving difference $T_j - T_m$, because its boundary temperature remains pinned at T_m regardless of state: it monopolises the fluid's heat, and segments downstream receive what is left.

Under Eq. (14) the same segment begins to superheat the instant its solid is exhausted. T_{pcm} rises, the driving difference $T_j - T_{\mathrm{pcm}}$ collapses, its demand falls away, and the heat continues downstream to segments that still have solid to melt. The mechanism is a negative feedback and it is automatic — nothing in the implementation distributes the heat, and no parameter was tuned.

Table 5: Local melted fraction at ten stations along the exchanger at end of charge, and its spread over all 100 segments.

formulation	min	max	spread
area-based (melted fraction capped)	0.595	1.000	0.405
enthalpy	1.000	1.000	**0.000**

The comparison in Table 5 is made at *equal exchanger size* and with the melted fraction capped in both formulations, so that only the sensible branches differ. This matters: an unguarded area-based formulation also produces $\varepsilon_{\mathrm{local}}$ above unity — up to 1.78 in this case — and quoting the resulting spread would conflate genuine non-uniformity with that over-melt artefact. The figures above isolate the effect of the sensible branches alone.

The physical reading is that PCM which never melts is dead volume, occupying the store and carrying no energy through the round trip. The enthalpy formulation finds substantially less of it.

Figure 4 shows why the summary statistics understate the effect. The area-based curve is not uniformly depressed; it is a *sawtooth* locked to the cascade. Each layer melts completely at its inlet end, where the fluid is 10 K above that layer's T_m, and incompletely at its outlet end, because segments upstream that have already finished continue to draw heat at the full driving difference and leave less for those behind. The enthalpy formulation removes the sawtooth rather than merely shifting its mean: a segment that finishes melting superheats, its own driving difference collapses, and the heat is passed along.

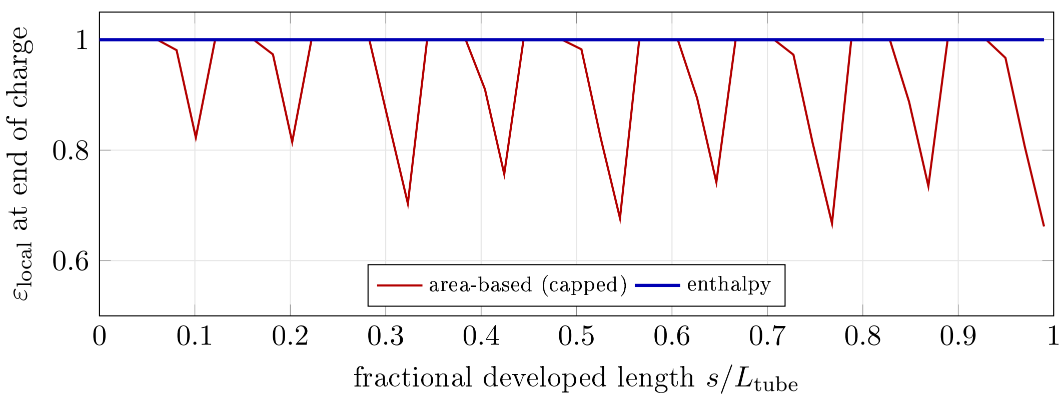

Figure 4: Local melted fraction along the exchanger at end of charge, compared at *equal* exchanger size with the melted fraction capped in both formulations. The sawtooth in the area-based curve is the cascade: each layer melts fully at its inlet end and incompletely at its outlet end, because a segment that has finished melting continues to draw heat at the full driving difference. Under the enthalpy formulation that segment superheats instead, its demand collapses, and the heat passes downstream — the curve is flat at unity. Spread falls from 0.405 to 0.000 (Table 5).

5.6 Cyclic steady state

The results above are *first-cycle* results: the store begins fully solid at T_m and the cycle does not return to that state. Repeating the charge–discharge schedule drives the store to a cyclic steady state (CSS) in which the end-of-cycle state reproduces the start-of-cycle state. Convergence takes of order thirty cycles here.

At CSS the enthalpy change over a complete cycle is zero by definition, and the store is adiabatic, so

$$\eta_{\text{storage}}(\text{CSS}) \;\equiv\; 1 \tag{20}$$

identically — for every exchanger size and every flow rate; computed values agree to eight decimal places. Equation (20) is energy conservation applied to a closed cycle rather than a result about the store, but it has a consequence for how such a model may be posed.

Sizing cannot be closed on the energy balance at CSS. A single-cycle model sizes the exchanger by requiring that it absorb a specified energy within the charging window. At CSS the absorbed and recovered energies are the *same number*, so that requirement and the corresponding discharge requirement collapse into one equation. It fixes the ratio of the two flow rates and says nothing about the exchanger size:

N_{wells}	flow ratio	charge/target	discharge/target
11.0	1.0740	1.00000	1.00000
14.0	1.0280	0.99994	0.99994
20.0	0.9965	0.99994	0.99994

Adding a loss mechanism would not remove this degeneracy: at CSS the charge would then exceed the discharge by exactly the loss, and the one remaining equation would still pin the flow ratio.

Reformulation. The resolution is to specify the hardware and *simulate* rather than solve. The exchanger count is taken from the latent inventory alone, $\Delta E/(\rho_m V_{\text{well}} h_m)$ — a deliberate overestimate, since it ignores the sensible capacity the store also has — and both flows are pinned by their specified glides. No quantity is solved for, so no convergence question arises, and the deviation of delivered energy from target becomes a reported result.

Table 6: Cyclic steady state: deviation from target against exchanger count. Each row is an independent march to CSS.

N_{wells}	delivered/target	ε end charge	ε end discharge
11.544	0.9426	1.000	0.154
14.000	0.9662	1.000	0.288
16.000	0.9777	1.000	0.376
20.000	0.9931	1.000	0.505
26.000	1.0082	1.000	0.629

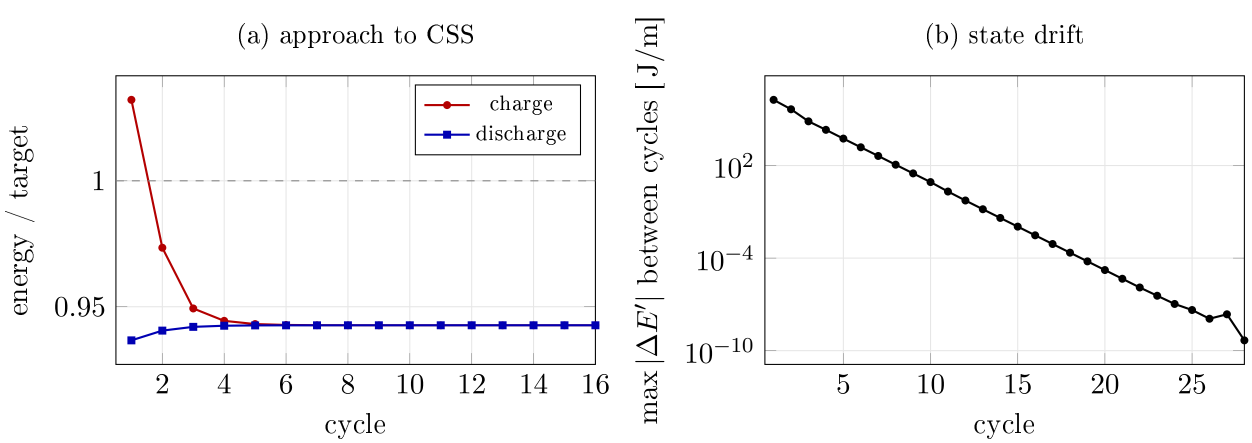

Figure 5: Convergence to cyclic steady state at the latent-only exchanger count. (a) The first cycle is the outlier: it starts from a cold, fully solid store and absorbs more than any later cycle. Charge and discharge converge to the *same* value, which is Eq. (20) in practice — the two curves become indistinguishable, not merely close. (b) The state drift falls geometrically; the run is stopped at 10^{-9}.

Figure 5(a) is worth dwelling on. The charge curve starts *above* target and falls; the discharge curve starts below and rises; they meet. That meeting is not a numerical coincidence but Eq. (20) asserting itself: once the state repeats, the two energies are the same number. The first cycle absorbs more than any later one because it alone begins from a cold, fully solid store — which is precisely why a first-cycle result cannot be quoted for a repeating duty.

At the latent-only size the field delivers 94.26 % of target. The store melts *completely* at end of charge — $\varepsilon_{\text{local}} = 1.000$ at every station — but returns only to a mean of 0.154, so the shortfall is a **discharge rate** limit rather than an inventory limit. The signature is in the last column, drawn in Fig. 6: the residual melt fraction *rises* with exchanger count, because each unit is then worked less hard and proportionally less of it refreezes. Were the limit an inventory limit it would fall, since a larger store would be proportionally emptier at the end of a fixed discharge. The two curves therefore separate the two failure modes without any additional calculation, and they place the lever on the discharge side — flow rate, window length, or conductance — rather than on more phase-change material.

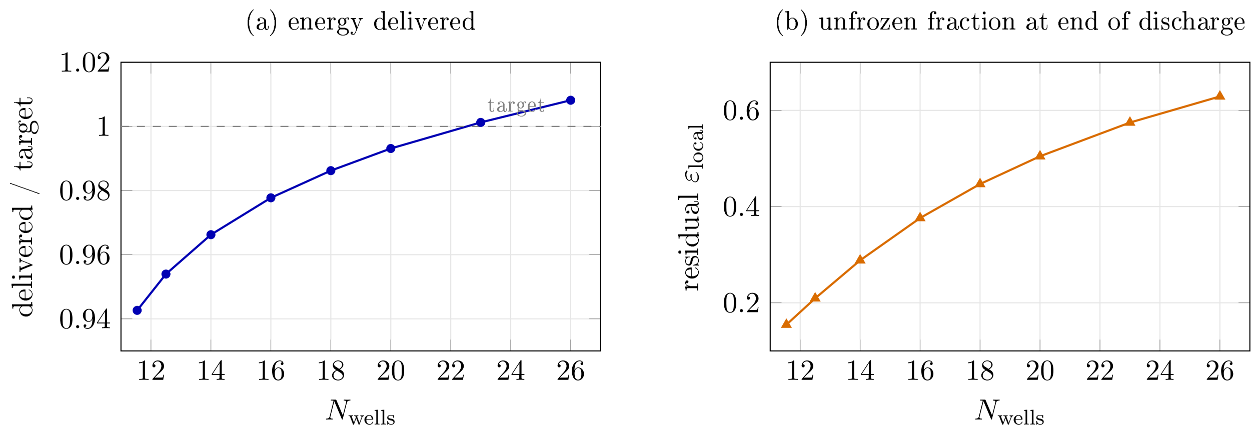

Figure 6: Cyclic steady state against exchanger count; each point is an independent march to convergence. (a) Delivered energy crosses target near $N_{\text{wells}} \approx 23$, about twice the latent-only count. (b) The diagnostic: the residual melted fraction at end of discharge *rises* with exchanger count, because each unit is then worked less hard and proportionally less of it refreezes. An inventory-limited store would show the opposite, since a larger store would be proportionally emptier after a fixed discharge. Note that the two panels carry different quantities on similar-looking curves; they are plotted separately for that reason.

5.7 Consequence for sizing

Because the formulation conserves energy and bounds the melted fraction, the store may be sized against two independent requirements rather than one. Writing ΔE for the energy to be stored, the field must both *absorb* it within the charging window and *contain* it at all:

$$N_{\text{wells}} = \max\bigl(N_{\text{rate}},\, N_{\text{capacity}}\bigr), \qquad N_{\text{capacity}} = \frac{\Delta E}{\rho_m V_{\text{well}}\,[h_m + c_{p,l}\Delta T]}. \tag{21}$$

N_{rate} follows from a root solve on the march and carries every thermal parameter; N_{capacity} is closed form and carries none, which makes it a hard lower bound no design can better. This applies to the *first* cycle; §5.6 shows that the rate criterion does not survive the passage to cyclic steady state.

6 Conclusions

A lumped enthalpy state has been formulated for phase-change material in the segments of a one-dimensional heat exchanger model, together with the closure that couples it to an effectiveness–NTU march and an exact time integrator for the resulting equation.

1. Carrying enthalpy rather than melted area removes a structural defect in reduced-order latent storage models. A segment whose phase change is complete has somewhere to put further heat, so none is discarded and the melted fraction is confined to $[0, 1]$ by the constitutive relation rather than by a guard. In the case examined, an area-based formulation discarded up to 148 % of the delivered heat during discharge.

2. The conductance closing the segment equation is $K = \dot{m}c_p(1 - e^{-\mathrm{NTU}})/\Delta z$ and not U_iP_i. The two agree only as $\mathrm{NTU} \to 0$, differ by 4 % to 33 % over the case studied, and only the former carries the capacity-rate limit that prevents heat flowing from cold to hot as the wall conductance improves.

3. Explicit time stepping is exact on the latent plateau and inadmissible off it, which is why area-based formulations never encountered the constraint. The branchwise-exact scheme of §4 is unconditionally stable, makes energy closure structural, and agrees with a 200000-substep reference to 3×10^{-13}.

4. Sensible capacity is self-levelling. Although it carries only about 4.7 % of the energy in the case studied, it redistributes the melting: compared at equal exchanger size, the spread in local melted fraction falls from 0.405 to 0.000 and dead volume falls correspondingly. The effect is first-order in the distribution while being small in the energy, because it acts through the driving temperature difference rather than through the energy stored.

5. At cyclic steady state the recovered-to-stored energy ratio is identically unity in an adiabatic store (§5.6), which collapses the charge and discharge energy requirements into a single equation. That equation fixes the ratio of the two flow rates and leaves the exchanger size undetermined, so a single-cycle sizing procedure does not carry over. Posing the problem as a simulation — specify the hardware, march to CSS, report the deviation from target — removes the difficulty rather than working around it, and turns the residual melt fraction into a diagnostic that distinguishes a rate limit from an inventory limit.

Limitations. Three assumptions bound the results above and are stated here rather than left implicit. Hypothesis H3 is *saturated* at the design point: the melt fronts of neighbouring tubes are in contact over 99 % of the exchanger, so the annular conduction resistance is evaluated at the limit of its validity and the residual solid — which occupies the corners between tubes, on a longer and narrower conduction path than an annulus represents — has its resistance understated. Hypothesis H6 assumes the cell boundary adiabatic, which for an axially graded store requires a physical partition between layers that are, at the inlet end of the case studied, 48.9 K apart; a slab estimate places the neglected cross-boundary leak at some 25 % of the useful duty, and it flows in the direction that erodes the grading. Finally, the time grid is not converged: $\varepsilon_{\mathrm{PCM}}$ moves -0.3 % between 40 and 320 time levels, one-sided. The first two are geometric questions about a cell's outer boundary and would be settled together by a two-dimensional conduction calculation on the real cross-section, which is the natural next step. None of the three affects the formulation or the integrator, which are the subject of this paper; all three affect the quantitative results of §5, which should accordingly be read as optimistic. In particular the -5.7 % cyclic-steady-state deviation of

§5.6 sits inside the band those three assumptions span, and is better read as evidence that the store is discharge-rate limited than as a calibrated shortfall.

Nomenclature

A_{melt}	melted cross-section per unit tube length	m^2
A_{avail}	PCM cross-section owned by one segment	m^2
A_{flow}	tube flow area	m^2
c_p	specific heat capacity	J/kg/K
C_s, C_l	sensible capacity per unit tube length	J/m/K
E'	PCM enthalpy per unit tube length	J/m
E'_{lat}	latent capacity per unit tube length	J/m
E'_{eq}	equilibrium enthalpy	J/m
h_m	latent heat of fusion	J/kg
h_e, h_i	outer, inner heat transfer coefficient	$\text{W/m}^2\text{/K}$
K	segment conductance per unit tube length	W/m/K
k_m, k_w	PCM, wall thermal conductivity	W/m/K
L_f, t_f, n_f	fin length, thickness, count	m, m,–
$\dot{m}$	mass flow rate	kg/s
NTU	number of transfer units per segment	–
P_T	finned outer perimeter	m
q'	heat rate per unit tube length	W/m
r_i, r_e	tube inner, outer radius	m
s	developed tube coordinate	m
Ste	Stefan number, $c_p\Delta T/h_m$	–
T	fluid temperature	K
T_m, T_{pcm}	melting, PCM temperature	K
U_i	overall coefficient on inner area	$\text{W/m}^2\text{/K}$
Δz	segment length	m
δ	melt-layer thickness	m
$\varepsilon_{\text{local}}$	melted fraction of one segment	–
η_f, η_o	fin, overall surface efficiency	–
ρ_m	PCM density	kg/m^3

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